

KEVIN LEVY

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TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib), SQL, Shell Scripting
- **Data Analysis:** Data Validation & Visualization, Time Series Analysis
- **Statistics & Machine Learning:** Frequentist & Bayesian Inference, MCMC Sampling, Supervised & Unsupervised Learning, Neural Networks
- **Infrastructure & Computing:** Linux, High-Performance Computing (HPC), Docker, Git/GitHub
- **Cloud Platforms:** Google Cloud Platform (GCP; Vertex AI, BigQuery, Looker, Cloud Storage)

PROFESSIONAL EXPERIENCE

Doctoral Researcher, Astrophysics — University of Melbourne, Melbourne, Australia Jun 2022 – Jun 2026

- **Pipeline Development:** Led the development of two scientific end-to-end Python pipelines:
 1. Reconstructed and statistically characterized the Universe's matter distribution from a new South Pole Telescope dataset, validating its quality for high-precision cosmological analysis;
 2. Extracted unbiased galaxy-cluster mass estimates from millimeter-wave data, tested on simulations and used to forecast performance for future surveys. The pipeline was later applied to survey data, achieving a ~2× improvement in measurement precision.
- **Statistical Methods:** Applied maximum-likelihood estimation, jackknife resampling, and chi-square/PTE statistics for model fitting, covariance estimation, and detection significance.
- **Validation & Robustness:** Designed and ran a comprehensive suite of systematic tests to identify potential biases and ensure robust, reproducible measurements.
- **Collaboration & Communication:** Worked within an international team on shared, version-controlled codebases and presented results at international conferences.
- **Publications:** [Levy et al. 2026](#), [Levy et al. 2023](#)

BI Developer — Delta Information Technologies, Mamer, Luxembourg Dec 2021 – May 2022

- **Data Engineering & Visualization:** Used SQL to build and maintain database views integrated with the company's ERP platform, developing dashboards and charts to track financial, buying, and selling KPIs across the client base
- **Stakeholder Management:** Worked directly with clients on-site and via calls/Zoom to gather requirements, customizing dashboards to match each client's specific needs and preferences.

INDEPENDENT PROJECTS

Flood Detection from Satellite Imagery

- Built an end-to-end remote sensing pipeline for flood detection by training a PyTorch U-Net from Sentinel-1 SAR imagery, achieving 79% mean IoU (segmentation accuracy).
- Deployed the model as a live, publicly callable API on Google Cloud Run, using a custom containerized training job on Vertex AI for scalable training.

Bayesian Model Selection for Galaxy Distance Distributions

- Built an automated Python pipeline fitting Gaussian Mixture Models to noisy galaxy distance distribution data via MCMC, using BIC for data-driven model-order selection.
- Designed an iterative reweighting scheme to jointly calibrate a bias correction across multiple data subsets.

EDUCATION

- Ph.D. in Astrophysics — University of Melbourne, Melbourne, Australia Jun 2022 – Jun 2026
- M.Sc. in Astrophysics — University of Bonn, Bonn, Germany Oct 2019 – Oct 2021
- B.Sc. in Physics — University of Bonn, Bonn, Germany Oct 2015 – Oct 2019

CERTIFICATIONS & AWARDS

- Google Cloud Computing Foundations Certificate — Google (2026)
- Data Scientist in Python Path — Dataquest.io (2021)
- N.D. Goldsworthy Scholarship (2022) — Awarded on the basis of academic excellence

LANGUAGES

Luxembourgish (Native) · English (C2) · German (C2) · French (B2)